



## **Chapter 2: Literature Review**

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**“Development of a comprehensive mobile application for export crop  
knowledge and trade facilitation”**

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## **1) Introduction**

Mobile applications have emerged as transformative tools in the agricultural sector, particularly for small-scale farmers in developing regions. These technologies address critical challenges such as limited market access, lack of real-time information, and inefficiencies in resource management. Existing studies emphasize the positive impacts of mobile solutions in bridging the digital divide, enhancing productivity, and improving livelihoods. However, most of this research focuses on local markets and general farming practices, with limited attention given to the specific needs of export-oriented agricultural sectors. Sri Lanka, known for its export crops such as tea, cinnamon, and coconut, requires tailored technological solutions that address global market dynamics, export compliance, and international standards. This literature review examines key studies on mobile applications in agriculture, highlighting their achievements, shortcomings, and the gaps that underscore the need for research targeting export crop industries. This analysis lays the groundwork for designing innovative tools to support Sri Lanka's agricultural exporters in navigating global challenges.

## **2) Literature review**

Applications, commonly referred to as "apps," are specialized software programs designed to perform specific functions for users. These apps operate on various digital platforms and have become integral to modern technology, serving diverse needs such as productivity, communication, and entertainment. The evolution of apps has led to their ubiquitous presence in personal, professional, and educational contexts.

Scholars have defined apps in various ways, emphasizing their functionality and purpose. Huang and Benyoucef (2015) described apps as "software applications that execute tasks tailored to user requirements, ranging from communication to entertainment." Similarly, Smith et al. (2018) defined them as "software programs that deliver services or content to users through networked devices." These definitions highlight the central role apps play in meeting user-specific needs through customizable and interactive interfaces.

Apps are often distinguished by their adaptability to different technological platforms, such as mobile devices and web browsers, further broadening their scope and usability. Whether

designed for leisure, information dissemination, or work efficiency, apps have become indispensable in simplifying complex tasks and fostering connectivity.

Apps are primarily classified into three major types: mobile apps, web-based apps, and hybrid apps. Each category serves unique user requirements and technological scenarios.

### 1. Mobile Apps

Mobile apps are software programs created specifically for smartphones and tablets. They are typically downloaded and installed via app marketplaces, such as the Google Play Store or Apple App Store. Mobile apps are valued for their portability and functionality, enabling users to perform various tasks, from managing schedules to engaging in e-commerce. Research by Johnson and Smith (2019) highlighted that mobile apps are designed for convenience, offering features like offline access and location-based services, making them ideal for on-the-go tasks. Examples of mobile apps include WhatsApp, fitness tracking apps, and mobile payment solutions.

### 2. Web-Based Apps

Web-based apps operate through web browsers and do not require users to download or install them. They are accessible from any device with an internet connection and offer greater flexibility and scalability compared to mobile apps. These apps are particularly suitable for tasks requiring frequent updates or large-scale user engagement. For instance, cloud-based platforms such as Google Docs or online learning systems like Coursera fall under this category. Brown et al. (2020) emphasized that web-based apps are advantageous for their cross-platform accessibility and dynamic features, making them popular for business and education purposes.

### 3. Hybrid Apps

Hybrid apps are a combination of mobile and web-based apps, leveraging the strengths of both. They are built using web technologies like HTML5 but are packaged in a way that allows them to function as native apps on mobile devices. Hybrid apps offer an optimal blend of performance and versatility. For example, Instagram is a hybrid app that provides a seamless user experience across different platforms.

The development of an application is a multi-phase process that transforms an idea into a functional and user-friendly digital product. It begins with requirement gathering and

planning, where developers identify the purpose of the application, its target audience, and the key features that need to be incorporated. This phase is critical for defining the scope of the project and involves conducting market research to understand user needs and industry trends. This approach ensures that the application will serve its intended purpose effectively. (Ratnaningrum et al., 2020)

The next stage is the design phase, which focuses on the application's layout and user interface. Developers and designers work together to create wireframes and prototypes to visualize the app's structure and interaction flow. This step often utilizes tools like Figma or Adobe XD to ensure a seamless and aesthetically pleasing user experience. It is at this point that user feedback can be incorporated to refine the design (Al-Razgan et al., 2020).

Once the design is approved, the process moves into the development phase. This stage involves actual coding and integrating functionalities into the app. The choice of programming language and framework depends on the platform, such as Swift for iOS applications or Kotlin for Android. Cross-platform frameworks like Flutter and React Native are also widely used for faster development (Banerjee et al., 2019). Developers often use modular programming approaches to ensure the code is reusable and maintainable.

After development comes the testing phase, which ensures that the app operates as intended. Quality assurance teams use manual and automated testing methods to identify and fix bugs. This stage also evaluates the application's performance, security, and usability. Tools like Selenium and Appium are widely used for automation, while real-time user testing ensures the app meets its intended purpose (Survanegara et al., 2020).

When testing is complete, the application is ready for deployment, where it is published on app distribution platforms such as Google Play Store or Apple App Store. This phase includes preparing promotional materials and adhering to the platform's publishing guidelines.

The final step in the process is maintenance and updates, where developers address user feedback, fix bugs, and implement new features to improve the app's functionality. Agile methodologies and feedback loops are commonly used to ensure the application evolves in line with user expectations and technological advancements (Srinivasan et al., 2018).

App development is a structured and iterative process that integrates planning, designing, developing, testing, and maintaining. Understanding these phases and their significance ensures the development of a high-quality application tailored to user needs.

## **2.1 Theoretical Review**

Qiang et al. (2012) investigated how mobile applications could revolutionize agriculture and rural development, especially in low-income regions with limited resources. The study highlights how mobile technologies have transformed traditional farming by providing better access to information, lowering transaction costs, and enabling market participation for small-scale farmers. Through examples like DrumNet in Kenya and AgriManagr, the research showcases how mobile platforms improve efficiency, transparency, and profitability in rural economies.

DrumNet, a key case study, connected farmers to markets and supply chains, boosting their incomes significantly. The platform integrated market information, credit services, and supply chain management into a single system, giving farmers the tools to optimize their production and marketing strategies. Similarly, AgriManagr streamlined tea production in Kenya by digitizing records and reducing transaction costs, helping farmers manage their activities more efficiently and meet market demands.

A major point in the study is how mobile applications address information asymmetry, which is a common issue for smallholder farmers. These farmers often lack timely and accurate data about market trends, prices, and demand. Mobile apps help bridge this gap by providing real-time information, enabling farmers to make informed decisions about planting, harvesting, and selling. They also bring transparency to transactions, reducing exploitation by middlemen and giving farmers more control over their sales.

Despite these benefits, the study highlights challenges such as low digital literacy, poor infrastructure, and the high cost of devices and services. These barriers limit the adoption of mobile technology in rural areas. To overcome these challenges, the researchers emphasize the need for user-friendly app designs and partnerships with governments, NGOs, and private companies. Investments in infrastructure and training are also crucial to ensure these technologies reach their full potential.

The findings are highly relevant to the development of a mobile application for Sri Lanka's agricultural export sector. Similar to the Kenyan farmers, Sri Lankan tea, cinnamon, and coconut exporters struggle with accessing real-time market data and connecting directly with buyers. Features such as supply chain management, financial tools, and market intelligence can

empower them to compete more effectively in global markets. Additionally, addressing local challenges like low internet access and limited digital literacy will be essential for the app's success and widespread use.

In conclusion, Qiang et al. (2012) demonstrate the transformative potential of mobile applications in agriculture. By tackling challenges and integrating diverse functionalities, these tools can promote sustainable development and improve livelihoods. Their insights offer a valuable roadmap for creating a mobile app tailored to Sri Lanka's agricultural export industry, helping farmers and exporters enhance their competitiveness and resilience.

Goltyapin (2014) delves into the potential of mobile applications to address several critical agricultural challenges, particularly for smallholder farmers. The study explores how mobile tools can enhance productivity, improve resource management, and facilitate better market access. By categorizing these applications into different functionalities—such as weather updates, pest control, market price monitoring, and supply chain management—the study highlights the growing role of mobile technology in modern agriculture. The use of these tools is particularly important for small-scale farmers, as they help bridge the gap in knowledge and market access that many rural farmers face.

One of the key strengths of mobile applications, as discussed in the study, is their versatility. Goltyapin emphasizes that these platforms are not only about providing information but also enable community-building among farmers. For instance, some mobile applications allow farmers to form virtual cooperatives, where they can share resources, best practices, and even negotiate collectively with buyers. This collaborative approach is significant as it empowers farmers, especially smallholders, by enhancing their bargaining power and enabling them to access better market opportunities. These applications also enable farmers to manage their resources more efficiently by providing real-time information on factors like weather, pest outbreaks, and soil conditions. By having such data at their fingertips, farmers can make informed decisions that ultimately help them reduce losses and increase productivity.

Moreover, the study highlights the importance of designing mobile applications with the user in mind. Farmers often have limited digital literacy, so the mobile tools need to be simple to use, with easy navigation and clear interfaces. Features like multilingual support and offline functionality are essential to ensuring that the tools are accessible to a broader audience. In Sri Lanka, for instance, an application targeting tea, cinnamon, and coconut farmers would need

to offer content in the local languages (Sinhala and Tamil), ensuring that it is relevant and easily understood by the farmers. This kind of user-centric design plays a vital role in the adoption and success of these technologies in rural communities.

However, the study also acknowledges the challenges that come with adopting mobile applications in agriculture. Issues such as the high cost of smartphones, limited awareness of the benefits of these tools, and resistance to change, particularly among older farmers, can hinder widespread adoption. To address these barriers, Goltypin suggests initiatives like targeted training programs to increase digital literacy and subsidies for mobile devices. Collaboration between local agricultural organizations and technology developers is also crucial for overcoming these challenges and promoting the adoption of mobile applications.

For my project on developing a mobile application for Sri Lanka's agricultural export sector, Goltypin's research provides important insights. Like the farmers studied in the research, Sri Lankan farmers in the tea, cinnamon, and coconut industries could greatly benefit from mobile platforms that offer real-time market updates, pest control advice, weather forecasts, and resource management tools. Additionally, by incorporating community-building features and offering training, the platform could empower farmers and improve their market access. This approach could help optimize their practices and contribute to better economic outcomes in the long run.

In conclusion, Goltypin (2014) highlights the transformative potential of mobile applications in agriculture, particularly for smallholder farmers. By focusing on user-centric design, overcoming adoption barriers, and integrating valuable resources like market intelligence and weather updates, mobile apps can significantly improve productivity and economic outcomes for farmers. These insights are essential in shaping a mobile application that meets the needs of Sri Lanka's agricultural export sector and helps farmers navigate the challenges of the global market.

Meena et al. (2018) explore the role of mobile applications in transforming the agricultural sector in India, particularly for smallholder farmers. Agriculture serves as the main source of livelihood for millions in India, yet many farmers face challenges such as fluctuating market prices, lack of reliable information, and limited access to resources. These issues often prevent farmers from achieving sustainable growth. In this context, mobile applications are seen as a



potential solution to address these gaps by offering real-time information and making it easier for farmers to access essential data for better decision-making.

One of the key strengths of mobile applications is their ability to provide real-time information tailored to the needs of farmers. According to Meena et al., mobile apps can offer updates on market prices, helping farmers identify the best times and places to sell their produce. In rural areas, where there is often limited access to market information, intermediaries tend to exploit this knowledge gap by offering lower prices. By giving farmers access to accurate market data, mobile applications can reduce the impact of intermediaries and ensure that farmers receive fairer prices. Additionally, weather forecasting features integrated into mobile apps can help farmers plan their agricultural activities more effectively, reducing losses caused by unpredictable weather.

The study also emphasizes the significance of advisory services within mobile applications. These services can provide farmers with expert guidance on critical agricultural activities, such as pest control, fertilizer application, and sustainable resource management. This expert advice can help farmers optimize their agricultural practices, resulting in increased productivity and improved crop quality. The authors stress that for these mobile applications to be effective, they must be user-friendly, support local languages, and function offline to ensure they are accessible to farmers in rural areas, many of whom may have limited digital literacy or unreliable internet connections.

Despite the advantages, Meena et al. identify several barriers to the widespread adoption of mobile applications among farmers. These barriers include the high cost of smartphones, limited technical knowledge among farmers, and inadequate digital infrastructure in rural areas. To address these challenges, the study suggests that governments, NGOs, and private sector partners collaborate to create a conducive environment for mobile technology adoption. This could involve offering subsidies, conducting training programs, and investing in infrastructure development.

The findings of this study are highly relevant to my project, which aims to develop a mobile application for Sri Lanka's agricultural export sector. Like the farmers in India, Sri Lankan farmers who grow crops like tea, cinnamon, and coconut often face similar challenges, such as limited market access and fluctuating prices. By integrating features such as real-time market updates, weather forecasting, and expert advisory services, my mobile application can

empower Sri Lankan farmers to make informed decisions and improve their profitability. Additionally, addressing barriers such as digital literacy and internet connectivity will be essential for ensuring that the application is widely adopted and has a lasting impact.

In conclusion, Meena et al. (2018) highlight the transformative potential of mobile applications in the agricultural sector. Their findings suggest that by providing real-time information, expert advice, and a user-friendly interface, mobile applications can help farmers improve productivity, access markets more effectively, and navigate challenges more efficiently. These insights can guide the development of a mobile platform for Sri Lanka's agricultural export sector, promoting greater competitiveness and sustainability in the global market.

Shams et al. (2020) examine the design of agriculture-focused mobile applications in Bangladesh through a value-sensitive lens. The study investigates 1,522 user reviews from mobile applications to understand how developers can better reflect user preferences in their designs. Schwartz's human values theory forms the theoretical foundation of the study, which identifies critical gaps between user expectations and the functionalities provided by existing applications. For instance, many applications prioritize productivity-enhancing features but fail to address underlying user needs, such as security, social connectivity, and conformity.

The research argues that value-sensitive design is crucial to the success of mobile applications, particularly in the context of agriculture. By aligning app features with users' cultural and personal values, developers can increase user engagement and satisfaction. While the study offers valuable insights, it remains theoretical and lacks empirical data to validate its recommendations in real-world scenarios. This limitation presents an opportunity for future research to test value-sensitive design principles in application development.

For Sri Lanka's agricultural export sector, this study underscores the importance of user-centric design when developing mobile applications for tea, cinnamon, and coconut exporters. By understanding the values and needs of local stakeholders, developers can create tools that resonate with users, improve adoption rates, and enhance overall effectiveness. Tailored solutions that incorporate localized content, multilingual support, and offline capabilities could significantly benefit small-scale exporters and foster sustainable growth in the sector.

Oteyo et al. (2021) provide an extensive survey of mobile applications designed for smart agriculture, emphasizing their potential to revolutionize farming practices through the

integration of advanced technologies like IoT and AI. The study categorizes mobile applications based on their functionalities, including soil monitoring, crop health detection, irrigation management, and market intelligence. The authors highlight that IoT-enabled applications can optimize resource usage, reduce waste, and increase yields by delivering real-time data and predictive insights.

One of the key contributions of this study is its discussion of the limitations and challenges associated with mobile applications for smart agriculture. These include high development and maintenance costs, the lack of robust digital infrastructure in rural areas, and the limited technical literacy of farmers. The authors advocate for collaborative efforts between governments, private sectors, and technology providers to address these barriers and make smart agriculture solutions more accessible.

Although the study does not provide empirical data, its theoretical insights are highly relevant to the development of mobile applications for Sri Lanka's export crop sector. Features such as predictive analytics, real-time alerts, and IoT integration could significantly enhance the productivity and competitiveness of tea, cinnamon, and coconut exporters. Moreover, the study highlights the importance of developing cost-effective and user-friendly solutions that cater to the unique needs of smallholder farmers in rural areas.

## **2.2 Empirical Review**

The study by Muñoz et al. (2020) focuses on creating the Market Mobile Application to support farmers and fishermen in the Philippines. These workers face significant challenges, such as poor market access, exploitation by middlemen, and a lack of up-to-date information needed for decision-making. The app was developed to address these issues by using mobile technology to bridge the gap caused by the digital divide in rural areas, helping farmers and fishermen improve their livelihoods and participate in economic activities more effectively.

A key feature of the Market Mobile Application is its ability to connect rural producers directly with buyers. By removing the need for intermediaries, the app ensures that farmers and fishermen receive fair prices for their products, leading to more stable incomes. In addition to this, the app includes various tools, such as weather alerts, directories of government resources,

communication features, and training modules. These tools not only boost productivity but also equip users with essential skills and information, enabling them to make informed decisions and access broader economic opportunities.

The researchers emphasized the importance of creating a user-friendly and inclusive technology. Many rural communities face challenges like unreliable internet access and low levels of digital literacy. To address this, the app includes offline functionality and a simple interface, ensuring that it remains accessible to users with minimal technological experience. This approach supports digital inclusion and helps rural workers use mobile technology as a tool for economic growth.

The study used both quantitative and qualitative methods to evaluate the app's effectiveness. Surveys revealed that users experienced significant improvements in their ability to negotiate better market terms, access vital services, and enhance their financial stability. Interviews and feedback from users also showed how the app encouraged community engagement by enabling people to share resources and knowledge, strengthening their economic resilience.

These findings are highly relevant to Sri Lanka, where farmers and exporters in the tea, cinnamon, and coconut sectors face similar challenges. Like their Filipino counterparts, small-scale Sri Lankan farmers often struggle with limited access to centralized, real-time market information and fair-trading opportunities. Developing a mobile app with features such as a digital marketplace, weather alerts, training tools, and government resources could help address these issues. To make the app effective, it would need to include offline functionality and a user-friendly design to overcome the connectivity and literacy barriers faced by rural users.

Overall, Muñoz et al. (2020) demonstrate that mobile applications can play a transformative role in empowering rural producers, promoting inclusivity, and supporting local industries. By applying similar strategies, Sri Lanka's export crop sector could enhance market access, competitiveness, and sustainability, benefiting small-scale farmers and the economy as a whole.

Saville et al. (2015) explored how mobile phones have transformed the lives of small-scale fishermen in Indonesia, focusing on their role in improving economic opportunities, safety, and community resilience. The study highlights how mobile technology can address key challenges

in traditional industries, making it particularly relevant to Sri Lanka's agricultural export sector.

One of the key findings of the study is the economic advantage mobile phones offer to fishermen. By providing real-time access to market data, such as pricing and demand trends, mobile phones enable fishermen to negotiate better deals and avoid losses from unsold stock. The devices also simplify coordination with buyers and transporters, reducing reliance on intermediaries. This direct communication helps fishermen achieve better profit margins and capitalize on market opportunities more effectively.

In addition to economic benefits, the study underscores how mobile phones enhance safety during fishing activities. Fishing is a risky occupation, and real-time communication plays a crucial role in managing emergencies. Mobile phones allow fishermen to receive timely weather updates and coordinate rescue operations when needed. This not only improves individual safety but also contributes to the overall well-being of fishing communities by reducing occupational risks.

The study also emphasizes the social impact of mobile phones, which goes beyond individual benefits to strengthen community ties. Fishermen use mobile devices to share resources, collaborate on fishing strategies, and exchange critical information. This fosters a sense of collective resilience and cooperation within their communities. For example, mobile phones help fishermen build social networks, which are essential for economic stability and mental well-being. These networks also support cooperative practices, such as pooling resources or sharing information about fishing zones, further boosting community resilience.

The findings of this study have direct relevance to Sri Lanka's agricultural export sector. Similar to the fishermen in Indonesia, Sri Lankan farmers and exporters of tea, cinnamon, and coconut face challenges in accessing real-time market information and communicating directly with buyers. Features such as live market updates, communication tools, and safety alerts could be incorporated into the proposed mobile application to address these issues. Furthermore, community-focused features like knowledge-sharing forums or collaborative tools could help strengthen ties among farmers and exporters, enhancing their resilience and productivity.

In summary, Saville et al. (2015) demonstrate the transformative potential of mobile technology in traditional industries. By addressing both economic and social challenges,

mobile phones have empowered marginalized communities and improved their quality of life. These insights can be applied to Sri Lanka's export crop industry, showing how mobile applications can boost efficiency, competitiveness, and sustainability in global markets.

Mbaya (2023) explores the impact of the Viazisoko Digital Platform, a mobile application designed to enhance market access for smallholder potato farmers in Kenya. This study focuses on addressing critical barriers faced by small-scale farmers, such as limited access to up-to-date market information, dependence on intermediaries, and vulnerability to fluctuating market conditions. The research emphasizes the role of mobile technology in empowering farmers by providing them with the tools they need to make better-informed decisions, improve productivity, and achieve better financial outcomes. The findings are particularly relevant to my project, as they highlight how similar technological solutions could benefit smallholder farmers in Sri Lanka's agricultural export sector, particularly those involved in tea, cinnamon, and coconut farming.

The Viazisoko platform is unique in its comprehensive approach, addressing multiple challenges that farmers face in the supply chain. By offering real-time pricing information, distribution network updates, and agricultural tips, the platform helps farmers optimize their production and marketing strategies. One of the standout features is its ability to reduce the reliance on middlemen, enabling farmers to directly negotiate with buyers and access more favorable market opportunities. This direct access to markets not only improves profit margins but also ensures that farmers receive fairer prices for their produce, promoting financial stability. In addition, the platform offers risk management tools such as weather forecasts, pest control advice, and soil quality updates. These features allow farmers to mitigate potential risks, minimize losses, and enhance the overall productivity of their farms.

Mbaya's study underscores the role of mobile technology in fostering more equitable and competitive markets. Traditionally, smallholder farmers have been disadvantaged due to limited access to market information, leaving them at the mercy of intermediaries who often exploit their lack of knowledge. Viazisoko addresses this issue by promoting transparency in transactions, ensuring that farmers have equal access to the information needed to make informed decisions. Moreover, the platform's emphasis on user-friendly design and localized content is essential for its success in rural areas, where digital literacy is often low, and access to technology may be limited. By creating an interface that is both simple to navigate and

tailored to the specific needs of local farmers, the platform ensures that even those with limited technological experience can benefit from its features.

The findings from this study are highly applicable to the development of my mobile application for Sri Lanka's agricultural export sector. Like the Kenyan potato farmers, Sri Lankan farmers often struggle with accessing real-time market data and fair-trading opportunities. Integrating features such as live market updates, weather forecasts, and agricultural advice into my app could significantly improve the efficiency and competitiveness of Sri Lankan farmers in the global market. Moreover, incorporating risk management tools such as pest control and soil quality updates would help address some of the ongoing challenges faced by exporters of tea, cinnamon, and coconut.

Additionally, ensuring the mobile application is designed with simplicity and accessibility in mind will be crucial for its widespread adoption in rural Sri Lanka. Many small-scale farmers in the country face challenges related to digital literacy and connectivity. Therefore, ensuring that the app is both user-friendly and functional in low-connectivity environments will be essential for maximizing its impact.

Shanadi et al. (2022) explored the development of a mobile application that aimed to provide Good Agricultural Practices (GAPs) for potato farmers in Sri Lanka. The study focused on challenges such as improper pesticide usage, poor soil fertility management, and inefficient post-harvest practices, which contribute to low yields and poor crop quality. The goal was to integrate scientific agricultural guidelines into a mobile platform to help bridge the gap between traditional farming methods and modern agricultural advancements.

The mobile application developed in the study offered farmers practical guidance on sustainable pest management, fertilizer optimization, and efficient planting techniques. These are key factors in improving productivity and reducing operational costs. In addition, the app provided real-time updates on weather conditions, pest outbreaks, and market trends, which helped farmers make informed decisions. This capability enabled them to align their farming practices with changing environmental and market conditions, reducing the risks associated with unpredictable weather and fluctuating market prices. The user-friendly design of the application ensured that even farmers with limited digital literacy could use the platform effectively, which was critical for its success.

The research demonstrated that the application contributed significantly to improving crop yields and quality by helping farmers adopt better farming practices. It also played a key role in enhancing the knowledge and skills of farmers, thus promoting long-term capacity building. By providing access to information and training on sustainable agricultural practices, the app empowered farmers to increase productivity, reduce reliance on expensive inputs, and contribute to environmental sustainability.

The findings from this study highlight the potential of mobile technology to improve agricultural productivity, especially for smallholder farmers. The research underscores the value of using mobile applications as tools for disseminating technical knowledge and fostering better agricultural practices. This is highly relevant to my project, as Sri Lanka's farmers of export crops such as tea, cinnamon, and coconut face similar challenges in accessing modern farming techniques and critical information.

By integrating GAPs for tea, cinnamon, and coconut into a mobile application, my project could significantly enhance the quality and consistency of Sri Lanka's export crops. The app would not only improve the quality of products but also ensure that Sri Lankan farmers are able to meet international standards, increasing the competitiveness of their exports in global markets. Furthermore, by offering features such as real-time market updates, weather forecasts, and agricultural tips, the application could help farmers make better decisions and improve their profitability.

In conclusion, Shanadi et al. (2022) illustrate how mobile applications can improve agricultural practices and promote sustainable farming. Their study shows the transformative potential of technology in enhancing agricultural productivity and long-term development. The insights from this research can provide valuable guidance in developing a mobile platform that addresses the specific needs of Sri Lanka's agricultural export sector, potentially improving crop quality, market access, and the competitiveness of Sri Lankan products in global markets.

Funmilayo and Ademola (2021) examine the transformative role of Internet of Things (IoT) technologies in agribusiness across developing countries, focusing on their application in Africa. The study highlights how IoT can address critical challenges such as inefficient resource management, low productivity, and limited market access. Practical examples include the use of drones for crop health monitoring, sensors for soil and irrigation management, and e-commerce platforms for selling agricultural produce.



The study provides empirical evidence of IoT's impact on rural economies, demonstrating its potential to enhance productivity, reduce waste, and improve profitability. Additionally, the authors discuss the importance of IoT in fostering youth engagement in agriculture, presenting it as a solution to unemployment and rural-urban migration. However, they also acknowledge significant barriers, such as the lack of digital infrastructure, high costs, and limited technical knowledge among farmers.

This research is directly applicable to Sri Lanka's agricultural export sector, where similar challenges exist. IoT-enabled solutions could help tea, cinnamon, and coconut exporters improve efficiency, meet global standards, and access international markets. By incorporating features such as real-time monitoring and automated compliance tools, Sri Lankan farmers can achieve higher productivity and profitability while addressing global trade requirements.

Pachiyappan et al. (2022) investigate the economic feasibility of protected cultivation methods for horticultural crops in Western India. The study highlights how advanced farming techniques, such as polyhouses and controlled environments, can increase productivity, improve crop quality, and ensure resource efficiency. Using regression models and economic analysis, the study demonstrates that protected cultivation is highly profitable for small-scale farmers growing high-value crops like tomatoes and capsicums.

The findings also reveal that protected cultivation reduces dependency on weather conditions and minimizes pest infestations, resulting in higher-quality produce and better market prices. However, the study identifies barriers such as high initial investment costs, lack of technical knowledge, and limited access to credit facilities. The authors recommend targeted policies and subsidies to encourage adoption among smallholder farmers.

For Sri Lanka's export crop sector, adopting protected cultivation methods could help farmers meet international standards and increase competitiveness. By integrating technology and financial support, farmers could achieve better yields, optimize resources, and access premium export markets.

Burchi et al. (2018) present an innovative ICT-based system architecture for managing greenhouses, integrating automation and sensor technologies. The study focuses on improving resource efficiency and crop yields by providing real-time data on environmental conditions.

Empirical tests demonstrate significant improvements in productivity and reduced water and energy usage.

This study is particularly relevant for high-value crops requiring precise management, as it showcases how ICT tools can optimize farming practices. For Sri Lanka's tea, cinnamon, and coconut exporters, adopting similar technologies could ensure consistent quality and compliance with global standards.

The reviewed literature highlights several advancements in mobile applications and ICT tools for agriculture, particularly in improving productivity, market access, and efficiency for smallholder farmers. However, critical gaps remain in addressing the specific needs of export-oriented agriculture for crops like tea, cinnamon, and coconut in Sri Lanka.

Theoretical studies, such as those by Shams et al. (2020) and Oteyo et al. (2021), provide insights into the potential of mobile applications to bridge information gaps and integrate advanced technologies like IoT and AI. However, these studies remain generalized, lacking application to export-specific challenges, such as compliance with international certifications or addressing global market dynamics.

Empirical studies, including Muñoz et al. (2020) and Funmilayo & Ademola (2021), demonstrate the effectiveness of ICT tools in improving resource management and market access. While these findings are valuable, they predominantly focus on local agricultural systems or regions like Africa and the Philippines, leaving a gap in the context of export crops. Similarly, Pachiyappan et al. (2022) and Burchi et al. (2018) emphasize resource efficiency and productivity through technological interventions, but they do not explore scalable solutions for small-scale exporters in developing countries like Sri Lanka.

Additionally, studies such as Shanadi et al. (2022) highlight the potential of mobile applications in disseminating agricultural guidelines but are limited to single crops and localized farming. This reveals a gap in addressing the broader complexities of export-oriented supply chains, including real-time market intelligence, predictive analytics, and tools for international trade facilitation.

Therefore, existing research largely overlooks the development of cost-effective, accessible mobile applications tailored to the needs of Sri Lanka's agricultural export sector. Key issues like export certification, predictive analytics for global trends, and real-time task coordination for farmers remain underexplored.

This study aims to fill these gaps by proposing a mobile application that integrates tools for global market intelligence, compliance with export standards, and sustainability, ensuring accessibility for small-scale farmers with limited digital literacy.

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